

KALPAK KANUNGO

Generative AI & Agentic AI Engineer

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PROFESSIONAL SUMMARY

Engineer with **3 years** building and supporting enterprise data and workflow systems in regulated healthcare, now specialized in **Generative AI** and **Agentic AI**. Designs **multi-agent systems** that orchestrate established **LLMs** in **Python** — **supervisor / specialist delegation**, **RAG** over vector stores, **tool-calling** via APIs, and confidence-gated **human-in-the-loop** approval — rather than training models from scratch. Combines a strong **data engineering** and diagnostic foundation (**Oracle SQL**, **ETL**, cross-system integration analysis) with an **audit-trail-first** instinct from **GxP / 21 CFR Part 11** environments that translates directly into traceable, governable AI systems.

CORE SKILLS

Languages	Python, SQL (Oracle), PowerShell
Agentic AI & LLMs	Claude, Google Gemini, OpenAI GPT, LLaMA, LangGraph, LangChain, Multi-Agent Orchestration, Supervisor / Specialist Patterns, ReAct, Planning & Reflection, Self-Correction (AI-reviewing-AI), Human-in-the-Loop Design, Prompt Engineering, Tool Calling
RAG & Retrieval	ChromaDB, Sentence-Transformers, Embedding Pipelines, Semantic Search, Context Grounding, Chunking
APIs & Backend	FastAPI, Flask, Slack Bolt, REST APIs, Google Drive / Sheets API, SQLite Checkpointing, ngrok
Cloud & LLM Platforms	AWS Bedrock, Microsoft Azure (Functions, API Management, Monitor), Azure OpenAI, AWS (Cloud Practitioner)
Data & Governance	Oracle SQL, Informatica ETL, Source-to-Target Reconciliation, Data Quality Monitoring, GxP / 21 CFR Part 11 Audit-Trail Design

AI & AGENTIC AI PROJECTS

DS Traceability Steward — Agentic Slack Copilot 2026

Python · LangGraph · Google Gemini · ChromaDB · Sentence-Transformers · Slack Bolt · FastAPI · Google Drive API · SQLite

- Built an end-to-end **agentic AI pipeline** that autonomously links **Slack** user stories to **Design Specification (DS)** IDs in a **GxP-regulated** complaint-handling document, eliminating manual traceability across **323 DS entries** spanning **26 configuration families**.
- Designed a **multi-agent LangGraph** pipeline — a matcher agent (**semantic search** via **ChromaDB + Sentence-Transformers**), a **Gemini**-powered drafting agent that generates house-style DS entries when no match exists, and a **self-correction reviewer agent** that critiques and revises drafts before human review (an **AI-reviewing-AI** quality gate).
- Implemented a durable **human-in-the-loop approval gate** using LangGraph's **interrupt / resume** pattern with **SQLite checkpointing**, pausing mid-execution for Slack **Approve / Edit / Reject** decisions and resuming from the exact checkpoint with no state loss across restarts.
- Engineered a table-aware **python-docx** parser with **Google Drive** write-back — inserting approved entries into the correct family section and uploading a new revision (Drive revision history as an immutable audit trail per **21 CFR Part 11**); exposed via **FastAPI + Slack Bolt** (slash commands, Block Kit modals, button callbacks) with **Google Sheets** append logging for concurrent-safe audit tracking.

Architecture: Slack Modal → FastAPI → LangGraph Pipeline → Semantic Match / Gemini Draft → Reviewer Agent → Human Gate → Drive Write-back → Sheets Log

PROFESSIONAL EXPERIENCE

Cognizant Technology Solutions Sep 2023 – Present

Developer / Consultant — Enterprise Quality & Data Systems (Healthcare & Life Sciences) Kolkata, India

- Built **Python** and **SQL** automation that replaced recurring manual production monitoring on a validated enterprise platform, **automating 13 monitoring processes** and surfacing integration and data-load failures before business impact.
- Diagnosed cross-system **SAP** integration failures via **Oracle SQL** against transaction and **JMS** audit tables, ran **Informatica ETL** loads with source-to-target reconciliation, and authored **DS / IQ / MP** validation documentation with full requirement-to-implementation traceability under **GxP / 21 CFR Part 11** — executing **ValGenesis** IQ / PIQ scripts through multiple go-lives.

CERTIFICATIONS

- Claude Certified Architect — Foundations (Anthropic)
- Microsoft Certified: Azure Fundamentals (AZ-900)
- AWS Certified Cloud Practitioner
- Cognizant Vibe Code Hackathon

EDUCATION

University of Engineering and Management, Kolkata Jul 2019 – Jun 2023

B.Tech, Computer Science — CGPA: 9.05

Julien Day School — ISC (83%), 2019 • ICSE (90%), 2017